

BATHRA GALISH S

Mechanical Engineering Student | HVAC & Thermal Systems | CAD & FEA
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Coimbatore, TN



EDUCATION

Government College of Technology, Coimbatore

2023 – 2027

Bachelor of Engineering – Mechanical Engineering | CGPA: 8.50 (up to 6th Semester)

DVD Higher Secondary School, Nagercoil

2022 – 2023

Higher Secondary (Class XII) – 86%

INTERNSHIP EXPERIENCE

Sky Breeze AC Solutions, Coimbatore

Jul – Aug 2025

Project Intern – HVAC Design & Installation

- Designed duct layouts and performed heat load calculations for residential and commercial HVAC systems.
- Participated in full project lifecycle: duct fabrication (sheet metal cutting, bending, joint prep), installation, testing, and commissioning across 4 on-site field projects.
- Designed and oversaw complete HVAC duct system for Aryabhavan Restaurant, Mettupalayam — including duct sizing, FAD/grille placement, and exhaust fan layout using AutoCAD.
- Assisted in 2 additional commercial installation projects; reviewed engineering drawings and was present during manufacturing and site installation phases.
- Developed practical understanding of site coordination, safety practices, and real-time problem solving.

MSME Technology Development Centre (PPDC), Coimbatore

Dec 2024 – Jan 2025

Training Intern – Revit MEP / BIM | 5th Semester

- Completed 10-week government-sponsored BIM training; created a complete MEP building model integrating HVAC, electrical, and plumbing systems using Revit MEP.
- Performed duct routing, AHU placement, supply/return air system design, and heat load calculations within Revit MEP.
- Developed electrical and plumbing layouts; practiced clash detection and inter-disciplinary coordination.

PROJECTS

Variable Fan Speed Control with Temperature Sensor | Revit MEP

2025 – 2026 | Sem 6

Tools: Revit MEP, Arduino IDE | Microcontroller: Arduino

- Designed a complete variable fan speed control system using an Arduino microcontroller and temperature sensor; programmed the control logic in Arduino IDE to automatically adjust fan speed based on real-time temperature readings.
- Modelled the system layout and duct integration in Revit MEP; implemented PWM-based fan speed control with sensor feedback loop for dynamic thermal response.

Robotic Arm – DEXTERM | CAD Design & Failure Analysis

2025 – 2026 | Sem 6

Skill Development Lab, GCT Coimbatore | Tools: Fusion 360, Creo

- Designed full 3D CAD model of a 6-DOF robotic arm (DEXTERM) in Fusion 360/Creo for the college Skill Development Lab.
- Completed the project through 2 iterative design-failure cycles; analysed failure modes and redesigned joints/links accordingly in Fusion 360.
- Project outcome used to establish a functional robotic arm demonstration unit for the Skill Development Lab.

Phase Change Materials (PCM) for Thermal Energy Storage – Literature Review

2025 | Sem 6

Skill Development Lab, GCT Coimbatore

- Conducted an in-depth literature review on PCMs (solid-liquid transition, latent heat storage) and their applications in building energy efficiency and HVAC systems.
- Reviewed 15+ research papers; synthesised findings on organic, inorganic, and eutectic PCMs. (Unpublished independent study)

Connecting Rod FEA – Fatigue & Durability Analysis

2025 | Sem 6

Tools: ANSYS Mechanical APDL

- Performed finite element analysis of a connecting rod in ANSYS Mechanical APDL to predict stress distribution, fatigue life, and failure zones under compressive, tensile, and inertia loading.
- Identified critical failure locations at fillets and pin bores; proposed geometric modifications to improve durability.

Aryabhavan Restaurant HVAC Duct – Design, Fabrication & Installation

2025 | Sem 4

Sky Breeze AC Solutions, Mettupalayam | Tools: AutoCAD

- Led end-to-end HVAC duct project for a commercial restaurant: designed duct layout with FAD sizing (6"x4" to 12"x10"), grille placement, exhaust fan selection (540 CFM), and cut openings in AutoCAD.
- Oversaw sheet metal fabrication and ceiling-mounted duct installation; coordinated with site team for structural integration.

BIM-Based MEP Building Plan – HVAC, Electrical & Plumbing

Dec 2024 | Sem 5

MSME-PPDC, Coimbatore | Tools: Revit MEP

- Developed a complete multi-discipline MEP model of a commercial building using Revit MEP, integrating HVAC ducts, electrical systems, and plumbing layouts in a coordinated BIM environment.
- Applied clash detection to resolve inter-discipline conflicts; extracted design schedules and drawing layouts.

Filter Dust Sensor Detection System for HVAC Ducts

2024 | Sem 3

Mini Project – GCT Coimbatore | Team of 4 (with seniors)

- Designed and implemented a sensor-based system to detect clogged/dirty filters in HVAC ducts and split AC units, improving system efficiency and indoor air quality.
- Integrated dust sensor with an alert/notification mechanism to prompt timely filter cleaning or replacement.

Electric Scissor Lift – Design Thinking & Mechanism Design

2024 | Sem 3

Design Thinking Course Project – GCT Coimbatore | Tools: Engineering Calculations, SCAMPER Method

- Applied the SCAMPER design thinking methodology to conceptualise an electrically-actuated scissor lift mechanism for lifting heavy HVAC components (95–150 kg) during installation, reducing the need for 5–9 manual labourers.
- Selected Mild Steel (UTS: 370–550 MPa, Yield: 250–350 MPa) for platform and arms; performed force analysis calculating total base load of 2452.5 N and plotted horizontal/vertical force distribution vs. angle.
- Designed platform dimensions (750×1950×106 mm) and arm cross-section (33×33×914 mm); integrated rack-and-pinion system with electric motor (gear/belt drive) for reliable lifting motion.

Occupancy Sensor System for Local Bus Transportation

2023 | Sem 2

Academic Project – GCT Coimbatore

- Designed an occupancy sensor system for local buses to monitor and report passenger load in real-time, aimed at improving public transport efficiency and crowd management.
- Conceptualised sensor placement, signal flow, and data reporting mechanism for a live bus environment.

LEADERSHIP & EXTRACURRICULAR

President – GCT ISHRAE Student Chapter

2025 – 2026

- Lead the ISHRAE student chapter; organise technical seminars, industry interaction sessions, and knowledge-sharing events on HVAC and thermal engineering topics.

Member – Rotaract GCT Club

2024 – 2025

- Participated in community outreach and team-building activities.

TECHNICAL SKILLS

CAD & Design: Creo Parametric (Intermediate), Fusion 360, AutoCAD

Simulation & FEA: ANSYS Mechanical APDL

BIM & MEP: Revit MEP – HVAC, Electrical & Plumbing Modelling, Heat Load Calculation

Fabrication: HVAC Duct Fabrication & Installation, Sheet Metal Work, 3D Printing (Ultimaker Cura)

Soft Skills: Technical Presentations, Team Leadership, Site Coordination, Research & Literature Review

CERTIFICATIONS & TRAINING

- Revit MEP / BIM Training – MSME Technology Development Centre (PPDC), Coimbatore (2024–25)
- Public Speaking for Non-Native English Speakers (Certified)
- HVAC System Design & Installation – Sky Breeze AC Solutions, Coimbatore (2025)